

Vijay K

Bachelor of Engineering
Computer Science and Engineering
Chennai

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OBJECTIVE

Computer Science undergraduate interested in Applied AI, graph-based machine learning, and intelligent recommendation systems. Experienced in building AI-powered applications using FastAPI, Neo4j, FAISS, Redis, Docker, and PyTorch Geometric. Seeking research-oriented opportunities to contribute to scalable AI systems, graph intelligence, and real-world machine learning applications.

EDUCATION

B.E. Computer Science and Engineering - Sri Sivasubramaniya Nadar College of Engineering - Chennai 2023 – 27
CGPA: 8.35
Higher Secondary - Lakshmi Matric Hr. Sec. School - Trichy 2022 – 23
Percentage: 97.8%

PROJECTS

SkillGraph: Workforce Ontology and Graph-Based Recommendation System

April 2026

Python, FastAPI, PostgreSQL, Neo4j, Docker, Redis, Celery, FAISS, Next.js



- Designed a workforce ontology and graph-based intelligence system to model semantic relationships between skills, roles, and career pathways.
- Investigated graph representation learning and similarity-based retrieval approaches using Neo4j, FAISS, and graph learning concepts through PyTorch Geometric.
- Developed a research prototype integrating graph databases, vector retrieval, and graph learning workflows while exploring semantic relationship modeling and recommendation approaches.

Habits: AI-Driven Behavioral Analysis Platform

December 2025

Flutter, Firebase Firestore, Gemini API



- Developed an AI-assisted behavioral analysis platform that generates productivity insights from user activity patterns using Gemini API.
- Designed user behavior tracking workflows and insight generation pipelines for long-term habit monitoring.
- Built responsive UI workflows with state management and real-time Firebase synchronization for seamless user experience.

OPEN SOURCE EXPERIENCE

Open Source Contributor | GirlScript Summer of Code (GSSoC) 2026

Remote

May 2026 – Present

- Contributing to open-source projects through GirlScript Summer of Code (GSSoC) 2026 by resolving issues, implementing features, and improving application functionality.
- Collaborating with project maintainers through GitHub issues, pull requests, code reviews, and version control best practices.
- Contributed feature enhancements and bug fixes while working within collaborative open-source development workflows.
- Successfully merged 2 pull requests involving feature enhancements, bug fixes, and project improvements while collaborating with maintainers.

TECHNICAL SKILLS

- Programming Languages:** Python, Java, JavaScript, SQL
- Machine Learning & AI:** PyTorch, PyTorch Geometric, Scikit-learn, Pandas, Graph Neural Networks, GraphSAGE, NetworkX, FAISS
- Data Preprocessing:** Data Preprocessing, Feature Engineering, Model Evaluation, Data Analysis
- Backend & AI Systems:** FastAPI, Redis, Celery, Docker, REST APIs
- Databases:** PostgreSQL, Neo4j, Firebase Firestore, MongoDB
- Development Tools:** Git, GitHub Actions, CI/CD Pipelines

HACKATHONS AND WORKSHOPS

- Blind Coding Contest – Runner Up** February, 2026
Secured Runner-Up position by solving algorithmic challenges under constrained blind-coding conditions.
- SAP HackFest 2025 – National Pre-Finalist** April, 2025
Advanced through 3 competitive evaluation rounds among national-level participating teams.
- Construction Cost Prediction Challenge – Ranked 80th** March, 2026
Ranked 80th among 500+ participants in a machine learning project focused on construction cost prediction and predictive analytics.

POSITIONS OF RESPONSIBILITY

- Head Volunteer** - Invente Paper Presentation 2025

MISCELLANEOUS

- LeetCode** - Solved 200+ DSA problems on LeetCode.

